

Mass/Spring/Damper Systems Part 1

EGR 345

GVSU

Outline

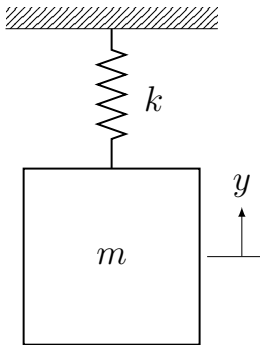
- review
- mass/spring/damper systems
 - initial condition response
 - Laplace with a second order denominator
 - Laplace with complex roots
 - critical damping
 - defining ζ (zeta)

Review: Two key characteristics of a dynamic system

- one “practical” and one mathematical
- vibrations are a transient response that can take a while to die out

Review: Mass/Spring Response

- what does this system do if you give it an initial displacement and release it?
- what differential equation describes this system's dynamics?



Review: What is a damper?

- what does a damper do?
- what does it represent?
- what equation models a damper?

Mass/Spring/Damper Response

- how does this system respond to an initial displacement?

